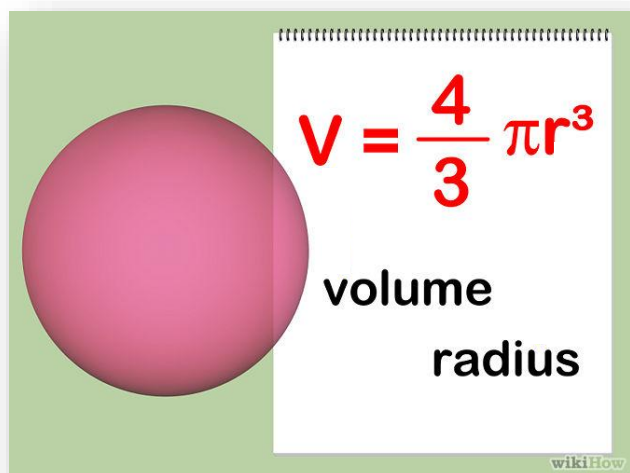


LESSON 8 : AREA AND VOLUME (B)

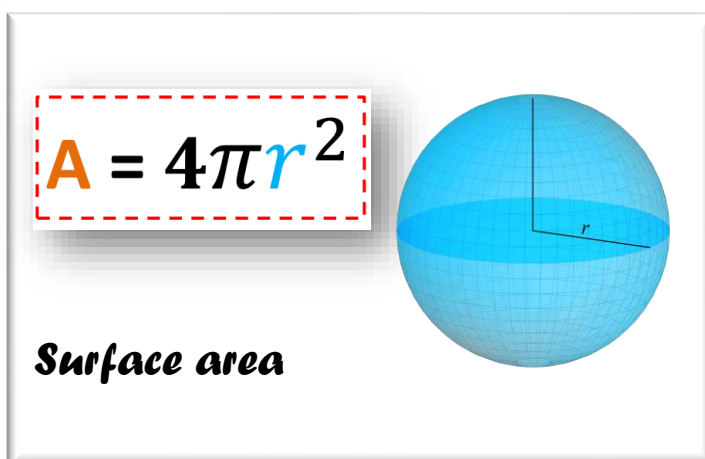
Checkpoint 1

Volume of Sphere



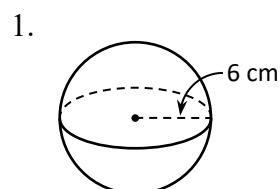
Checkpoint 2

Surface Area of Sphere

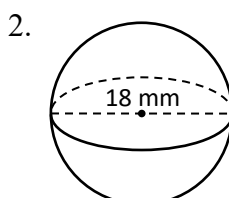


For each sphere or hemisphere in Q1–3, find the

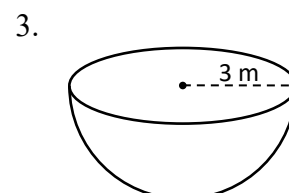
(a) volume, (b) total surface area. (Give the answers in terms of π .)



(a) Volume = $\frac{4}{3} \times \pi \times 6^3 \text{ cm}^3$
 $= 288\pi \text{ cm}^3$
 (b) Total surface area = $4 \times \pi \times 6^2 \text{ cm}^2$
 $= 144\pi \text{ cm}^2$

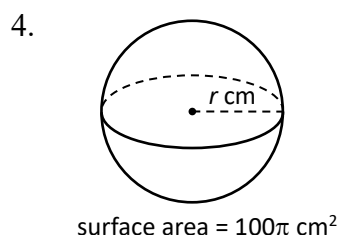


(a) Volume = $\frac{4}{3} \times \pi \times \left(\frac{18}{2}\right)^3 \text{ mm}^3$
 $= 972\pi \text{ mm}^3$
 (b) Total surface area = $4 \times \pi \times \left(\frac{18}{2}\right)^2 \text{ mm}^2$
 $= 324\pi \text{ mm}^2$

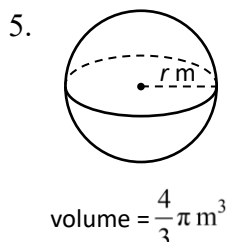


(a) Volume = $\frac{1}{2} \times \frac{4}{3} \times \pi \times 3^3 \text{ m}^3$
 $= 18\pi \text{ m}^3$
 (b) Total surface area
 = curved surface area + area of circular plane
 $= \left(\frac{1}{2} \times 4 \times \pi \times 3^2 + \pi \times 3^2\right) \text{ m}^2$
 $= 27\pi \text{ m}^2$

Find the unknown in each of the following spheres or hemispheres in Q4–6.



$4\pi r^2 = 100\pi$
 $r^2 = 25$
 $r = 5$



$\frac{4}{3} \pi r^3 = \frac{4}{3} \pi$
 $r^3 = 1$
 $r = 1$



total surface area = $12\pi \text{ cm}^2$

Total surface area
 = curved surface area + area of circular plane
 $= \left(\frac{1}{2} \times 4\pi r^2 + \pi r^2\right) \text{ cm}^2$
 $= 3\pi r^2 \text{ cm}^2$
 $\therefore 3\pi r^2 = 12\pi$
 $r^2 = 4$
 $r = 2$

PRE S3 MATHS L8 – AREA AND VOLUME (B)

7. The surface area of a spherical basketball is $1\,850\text{ cm}^2$. Find
- its radius,
 - its volume.
- (Give the answers correct to 3 significant figures.)

(a) Let $r\text{ cm}$ be the radius of the basketball.

$$4\pi r^2 = 1\,850$$

$$r^2 = \frac{925}{2\pi}$$

$$r = \sqrt{\frac{925}{2\pi}}$$

$$= 12.1, \text{ cor. to 3 sig. fig.}$$

$\therefore$ The radius of the basketball is 12.1 cm .

(b) Volume $= \frac{4}{3} \times \pi \times \left(\sqrt{\frac{925}{2\pi}}\right)^3\text{ cm}^3$

$$= \underline{7\,480\text{ cm}^3}, \text{ cor. to 3 sig. fig.}$$

8. The figure shows a hemisphere. The area of its base is $900\pi\text{ mm}^2$. Find its volume in terms of π .

Let $r\text{ mm}$ be the radius of the hemisphere.

$$\pi r^2 = 900\pi$$

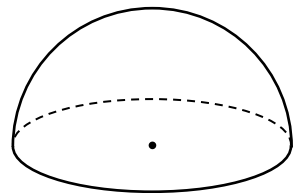
$$r^2 = 900$$

$$r = 30$$

$\therefore$ Volume of the hemisphere

$$= \frac{1}{2} \times \frac{4}{3} \times \pi \times 30^3\text{ mm}^3$$

$$= \underline{18\,000\pi\text{ mm}^3}$$



9. The diameter of a spherical Christmas ball is 7 cm . Find the surface area of the ball in terms of π .

$$\text{Surface area} = 4 \times \pi \times \left(\frac{7}{2}\right)^2\text{ cm}^2$$

$$= \underline{49\pi\text{ cm}^2}$$

10. The curved surface area of a hemisphere is 200 m^2 . Find the circumference of its base, correct to 3 significant figures.

Let $x\text{ m}$ be the radius of the hemisphere.

$$\frac{1}{2} \times 4\pi x^2 = 200$$

$$x^2 = \frac{100}{\pi}$$

$$x = \sqrt{\frac{100}{\pi}}$$

Circumference of its base

$$= 2\pi \times \sqrt{\frac{100}{\pi}}\text{ m}$$

$$= \underline{35.4\text{ m}}, \text{ cor. to 3 sig. fig.}$$

PRE S3 MATHS L8 – AREA AND VOLUME (B)

11. A large solid metal sphere of radius 15 cm is melted and recast into 27 identical small solid metal spheres. Find
- the radius of each small sphere,
 - the surface area of each small sphere in terms of π .

(a) Let r cm be the radius of each small sphere.

$\therefore$ Volume of the large sphere
= total volume of the 27 small spheres

$$\therefore \frac{4}{3} \times \pi \times 15^3 = 27 \times \frac{4}{3} \times \pi \times r^3$$

$$r^3 = 125$$

$$r = 5$$

$\therefore$ The radius of each small sphere is 5 cm.

(b) Surface area of each small sphere

$$= 4 \times \pi \times 5^2 \text{ cm}^2$$

$$= \underline{100\pi \text{ cm}^2}$$

12. The volumes of the sphere and the cylinder in the figure are the same. Find the base radius of the cylinder.

Let r cm be the base radius of the cylinder.

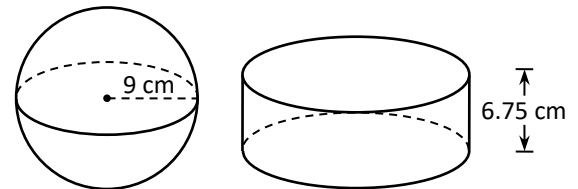
Volume of the sphere = volume of the cylinder

$$\frac{4}{3} \times \pi \times 9^3 = \pi \times r^2 \times 6.75$$

$$r^2 = 144$$

$$r = 12$$

$\therefore$ The base radius of the cylinder is 12 cm.



13. The figure shows a steel bowl in the shape of a hemisphere. The bowl is 0.2 cm thick with an internal diameter of 15.6 cm. If there is 600 cm^3 of steel, at most how many bowls can be made?

$$\text{Internal radius} = \frac{1}{2} \times 15.6 \text{ cm} = 7.8 \text{ cm}$$

$$\text{External radius} = (7.8 + 0.2) \text{ cm} = 8 \text{ cm}$$

Volume of steel for each bowl

$$= \left[\frac{1}{2} \times \frac{4}{3} \times \pi \times 8^3 - \frac{1}{2} \times \frac{4}{3} \times \pi \times 7.8^3 \right] \text{ cm}^3$$

$$= 78.431 \text{ cm}^3, \text{ cor. to 5 sig. fig.}$$

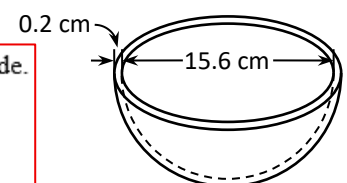
Let n be the number of bowls that can be made.

$$n \times 78.431 = 600$$

$$n = \frac{600}{78.431}$$

$$= 7.65, \text{ cor. to 3 sig. fig.}$$

$\therefore$ At most 7 bowls can be made.



PRE S3 MATHS L8 – AREA AND VOLUME (B)

14. In the figure, there is some water in a cylindrical container of base radius 4.8 cm. An iron sphere of diameter 7.2 cm is immersed in water. Then, the depth of water rises to 13 cm. Find the original depth of water.

Let h cm be the rise in water level.

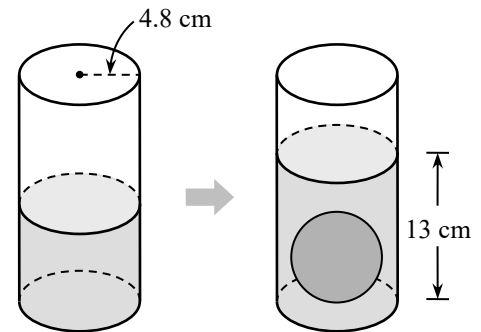
$\therefore$ Volume of the water that has risen
= volume of the iron sphere

$$\therefore \pi \times 4.8^2 \times h = \frac{4}{3} \times \pi \times \left(\frac{7.2}{2}\right)^3$$

$$23.04\pi h = 62.208\pi$$

$$h = 2.7$$

$\therefore$ Original depth of water = $(13 - 2.7)$ cm
= 10.3 cm



15. In the figure, a vessel consists of a hemisphere and a cylinder with a common base. The base diameter of the hemisphere is 30 cm. The capacity of the vessel is $9\,000\pi$ cm³. Is the height of the vessel less than 40 cm? Explain your answer.

Let h cm be the height of the cylinder.

Capacity of the vessel
= volume of the hemisphere + volume of the cylinder

$$= \left[\frac{1}{2} \times \frac{4}{3} \times \pi \times \left(\frac{30}{2}\right)^3 + \pi \times \left(\frac{30}{2}\right)^2 \times h \right] \text{ cm}^3$$

$$= (2\,250\pi + 225\pi h) \text{ cm}^3$$

$\therefore 2\,250\pi + 225\pi h = 9\,000\pi$

$$225\pi h = 6\,750\pi$$

$$h = 30$$

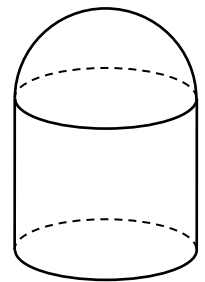
$\therefore$ The height of the vessel
= the height of the cylinder + the radius of the hemisphere

$$= \left(30 + \frac{30}{2} \right) \text{ cm}$$

$$= 45 \text{ cm}$$

$$> 40 \text{ cm}$$

$\therefore$ The height of the vessel is not less than 40 cm.



16. The solid in the figure is formed by two hemispheres. The radius of the smaller hemisphere is 4 cm. The curved surface area of the larger hemisphere is 40.5π cm². Find the total surface area of the solid, correct to the nearest cm².

Let r cm be the radius of the larger hemisphere.

$$\frac{1}{2} \times 4\pi r^2 = 40.5\pi$$

$$r^2 = 20.25$$

$$r = 4.5$$

Curved surface area of the smaller hemisphere

$$= \frac{1}{2} \times 4 \times \pi \times 4^2 \text{ cm}^2$$

$$= 32\pi \text{ cm}^2$$

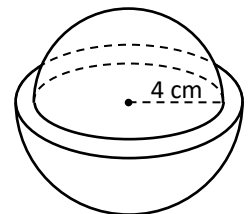
Area of the ring = $(\pi \times 4.5^2 - \pi \times 4^2) \text{ cm}^2$

$$= 4.25\pi \text{ cm}^2$$

Total surface area
= curved surface area of the smaller hemisphere +
curved surface area of the larger hemisphere +
area of the ring

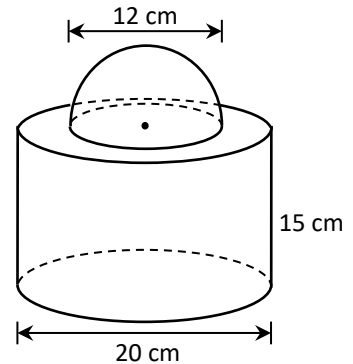
$$= (32\pi + 40.5\pi + 4.25\pi) \text{ cm}^2$$

$$= \underline{241 \text{ cm}^2}, \text{ cor. to the nearest cm}^2$$



PRE S3 MATHS L8 – AREA AND VOLUME (B)

17. The solid in the figure is formed by a right cylinder and a hemisphere. The base diameter and the height of the cylinder are 20 cm and 15 cm respectively. The base diameter of the hemisphere is 12 cm. Find
- the curved surface area of the cylinder,
 - the curved surface area of the hemisphere,
 - the total surface area of the solid.
- (Give the answers in terms of π .)



(a) Curved surface area of the cylinder

$$= \pi \times 20 \times 15 \text{ cm}^2$$

$$= \underline{300\pi \text{ cm}^2}$$

(b) Curved surface area of the hemisphere

$$= \frac{1}{2} \times 4 \times \pi \times \left(\frac{12}{2}\right)^2 \text{ cm}^2$$

$$= \underline{72\pi \text{ cm}^2}$$

(c) Total surface area of the solid

$$= [\pi \times 10^2 + (\pi \times 10^2 - \pi \times 6^2) + 300\pi + 72\pi] \text{ cm}^2$$

$$= \underline{536\pi \text{ cm}^2}$$

18. The figure shows a right circular cone of base radius 14.4 cm and a hemisphere of radius 12 cm. Their total surface areas are the same.
- Find the slant height of the cone.
 - Is the volume of the hemisphere greater than that of the cone? Explain your answer.

(a) Let ℓ cm be the slant height of the cone.

Total surface area of the cone

$$= (\pi \times 14.4 \times \ell + \pi \times 14.4^2) \text{ cm}^2$$

$$= (14.4\pi \ell + 207.36\pi) \text{ cm}^2$$

Total surface area of the hemisphere

$$= \left(\frac{1}{2} \times 4 \times \pi \times 12^2 + \pi \times 12^2 \right) \text{ cm}^2$$

$$= 432\pi \text{ cm}^2$$

$$14.4\pi \ell + 207.36\pi = 432\pi$$

$$14.4\pi \ell = 224.64\pi$$

$$\ell = 15.6$$

$\therefore$ The slant height of the cone is 15.6 cm.

(b) Height of the cone $= \sqrt{15.6^2 - 14.4^2}$ cm

$$= 6 \text{ cm}$$

Volume of the cone $= \frac{1}{3} \times \pi \times 14.4^2 \times 6 \text{ cm}^3$

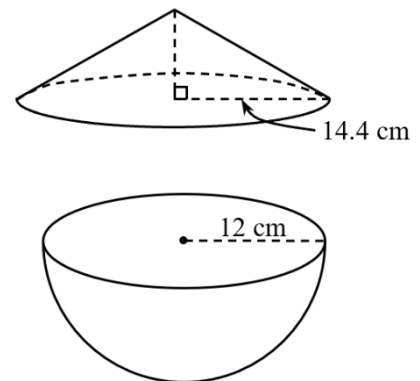
$$= 414.72\pi \text{ cm}^3$$

Volume of the hemisphere $= \frac{1}{2} \times \frac{4}{3} \times \pi \times 12^3 \text{ cm}^3$

$$= 1152\pi \text{ cm}^3$$

$$> 414.72\pi \text{ cm}^3$$

$\therefore$ The volume of the hemisphere is greater than that of the cone.



PRE S3 MATHS L8 – AREA AND VOLUME (B)

19. The figure shows a sausage, which is in the shape of a right cylinder with two hemispheres at both ends. The diameter of each hemisphere is 1.5 cm. The total surface area of the sausage is $20.25\pi \text{ cm}^2$.
- Find the length of the whole sausage.
 - Find the volume of the whole sausage in terms of π .



(a) Let x cm be the height of the cylinder.

$\therefore$ Total surface area of the sausage

$= 2 \times \text{curved surface area of each hemisphere} + \text{curved surface area of the cylinder}$

$= \text{surface area of the sphere} + \text{curved surface area of the cylinder}$

$$\therefore 4\pi \times \left(\frac{1.5}{2}\right)^2 + \pi \times 1.5 \times x = 20.25\pi$$

$$2.25 + 1.5x = 20.25$$

$$1.5x = 18$$

$$x = 12$$

$$\therefore \text{Length of the whole sausage} = \left(\frac{1.5}{2} + 12 + \frac{1.5}{2}\right) \text{ cm}$$

$$= \underline{13.5 \text{ cm}}$$

(b) Volume of the whole sausage

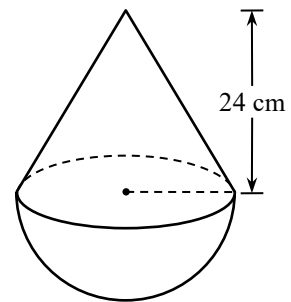
$= 2 \times \text{volume of each hemisphere} + \text{volume of the cylinder}$

$= \text{volume of the sphere} + \text{volume of the cylinder}$

$$= \left[\frac{4}{3} \times \pi \times \left(\frac{1.5}{2}\right)^3 + \pi \times \left(\frac{1.5}{2}\right)^2 \times 12 \right] \text{ cm}^3$$

$$= \underline{7.3125\pi \text{ cm}^3}$$

20. The solid in the figure consists of a right circular cone and a hemisphere with a common base. The height of the cone is 24 cm. The ratio of the volume of the cone to the volume of the hemisphere is 2 : 3.
- Find the radius of the hemisphere.
 - Find the volume of the solid in terms of π .
 - Find the total surface area of the solid in terms of π .



(a) Let r cm be the radius of the hemisphere.

$$\frac{\text{volume of the cone}}{\text{volume of the hemisphere}} = \frac{2}{3}$$

$$\frac{\frac{1}{3} \times \pi r^2 \times 24}{\frac{1}{2} \times \frac{4}{3} \pi r^3} = \frac{2}{3}$$

$$\frac{12}{r} = \frac{2}{3}$$

$$r = 18$$

$\therefore$ The radius of the hemisphere is 18 cm.

(b) Volume of the solid

$$= \left(\frac{1}{3} \times \pi \times 18^2 \times 24 + \frac{1}{2} \times \frac{4}{3} \times \pi \times 18^3 \right) \text{ cm}^3$$

$$= (2592\pi + 3888\pi) \text{ cm}^3$$

$$= \underline{6480\pi \text{ cm}^3}$$

(c) Let ℓ cm be the slant height of the cone.

$$\ell = \sqrt{18^2 + 24^2}$$

$$= 30$$

$\therefore$ Total surface area of the solid

$$= \left(\pi \times 18 \times 30 + \frac{1}{2} \times 4 \times \pi \times 18^2 \right) \text{ cm}^2$$

$$= (540\pi + 648\pi) \text{ cm}^2$$

$$= \underline{1188\pi \text{ cm}^2}$$

PRE S3 MATHS L8 – AREA AND VOLUME (B)

21. In the figure, a right cylindrical cup of base radius 6 cm and height 15 cm contains some water. When three spherical stones of radius 3 cm each are immersed in the water inside the cup, the water level rises to 14 cm.

- (a) Find the original water level.
(b) If the three stones are replaced by a new spherical stone of radius 4.5 cm and it is immersed in water, will the water overflow? Explain your answer.

(a) Let h cm be the original water level.

Total volume of the three stones = volume of water that has risen

$$3 \times \left(\frac{4}{3} \times \pi \times 3^3 \right) = \pi \times 6^2 \times (14 - h)$$

$$3 = 14 - h$$

$$h = 11$$

$\therefore$ The original water level is 11 cm.

(b) Total volume of water and the new stone

$$= \left(\pi \times 6^2 \times 11 + \frac{4}{3} \times \pi \times 4.5^3 \right) \text{ cm}^3$$

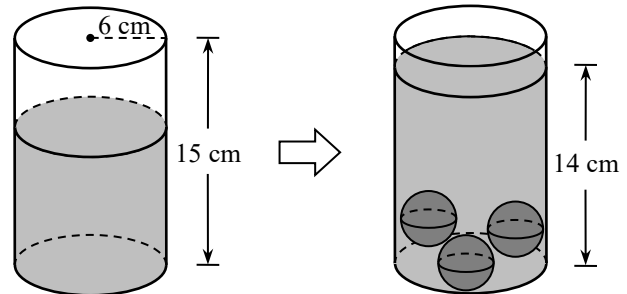
$$= 517.5\pi \text{ cm}^3$$

$$\text{Capacity of the cup} = \pi \times 6^2 \times 15 \text{ cm}^3$$

$$= 540\pi \text{ cm}^3$$

$$\therefore 517.5\pi < 540\pi$$

$\therefore$ The water will not overflow.



22. In the figure, a right circular cone of base radius 4 cm and height 8 cm is crafted from a wooden sphere.

- (a) Find the radius of the sphere.
(b) What percentage of the volume of the sphere is the volume of the cone?

(a) Refer to the notations in the figure.

Let r cm be the radius of the sphere, then

$OA = OC = r$ cm and $OB = (8 - r)$ cm.

In $\triangle OBC$,

$$OB^2 + BC^2 = OC^2 \quad (\text{Pyth. theorem})$$

$$(8 - r)^2 + 4^2 = r^2$$

$$64 - 16r + r^2 + 16 = r^2$$

$$16r = 80$$

$$r = 5$$

$\therefore$ The radius of the sphere is 5 cm.

(b) Volume of the cone

$$= \frac{1}{3} \times \pi \times 4^2 \times 8 \text{ cm}^3$$

$$= \frac{128}{3} \pi \text{ cm}^3$$

Volume of the sphere

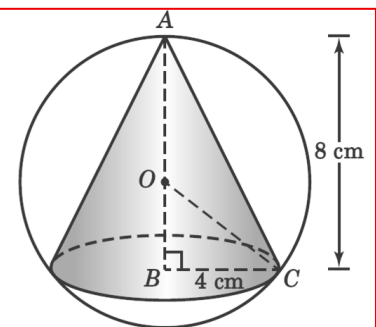
$$= \frac{4}{3} \times \pi \times 5^3 \text{ cm}^3$$

$$= \frac{500}{3} \pi \text{ cm}^3$$

The required percentage

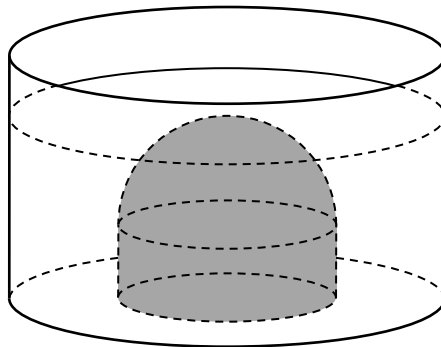
$$= \frac{\frac{128}{3} \pi}{\frac{500}{3} \pi} \times 100\%$$

$$= \underline{\underline{25.6\%}}$$



PRE S3 MATHS L8 – AREA AND VOLUME (B)

23. In the figure, a solid composed of a right cylinder and a hemisphere with the same base radius r is put into a cylindrical tank. The water is added to the tank such that it just covers the top of the solid. It is given that the base radius of the cylindrical tank is $2r$.



The solid is then taken out from the tank and the water level drops by 20%.

- (a) Express the height of the right cylinder in terms of r .
 (b) If the total surface area of the solid is $975\pi \text{ cm}^2$, express the volume of water in the tank in terms of π .

Let h be the height of the right cylinder.

(a) Drop in water level $= (r + h) \times 20\%$

$$= \frac{1}{5}r + \frac{1}{5}h$$

$$\therefore \pi \times (2r)^2 \times \left(\frac{1}{5}r + \frac{1}{5}h \right) = \frac{1}{2} \times \frac{4}{3} \pi r^3 + \pi r^2 h$$

$$\frac{4}{5} \pi r^3 + \frac{4}{5} \pi r^2 h = \frac{2}{3} \pi r^3 + \pi r^2 h$$

$$h = \frac{2}{3}r$$

$$\therefore \text{The height of the right cylinder} = \underline{\underline{\frac{2}{3}r}}$$

(b) Total surface area of the solid $= 975\pi \text{ cm}^2$

$$\frac{1}{2} \times 4\pi r^2 + 2\pi r h + \pi r^2 = 975\pi \text{ cm}^2$$

$$3\pi r^2 + 2\pi r \times \frac{2}{3}r = 975\pi \text{ cm}^2$$

$$\frac{13}{3} \pi r^2 = 975\pi \text{ cm}^2$$

$$r = 15 \text{ cm}$$

$\therefore$ Volume of water in the tank

$$= \pi \times (2r)^2 \times \left(\frac{4}{5}r + \frac{4}{5}h \right)$$

$$= \pi \times (2 \times 15)^2 \times \left(\frac{4}{5} \times 15 + \frac{4}{5} \times \frac{2}{3} \times 15 \right) \text{ cm}^3$$

$$= \underline{\underline{18\,000\pi \text{ cm}^3}}$$